

Element A

Identify Problem:

Special-needs kids at Whitemarsh Elementary School are in need of a new and improved sensory space, and are in need of new sensory devices. Sensory play is vital for the development of fine motor skills in children, and also helps to provide a calming environment in stressful situations at school. Additionally, the visual sense is one of the most important senses when it comes to education and development, and it's integral to learning how to read, write, play sports, etc.

Define Problem:

The PW engineers must design, test, and fabricate a new visual sensory device that will be implemented into the existing sensory room in order to provide stimulation, interactivity, and a calming experience, which will provide a better environment for the kids and teachers in the classroom. The PW engineers must collaborate with other PW engineering groups and teachers at Whitemarsh Elementary to figure out how to effectively implement this device into the sensory room. Additionally, the new device must be tested by the kids and teachers before it is finalized and put into the sensory space.

Stakeholders:

Primary: Students in the classroom who would directly benefit from a newer and better visual sensory project. Also, teachers whose job might be easier because of an improved overall sensory area.

Secondary: Parents of students whose improved in-school experience may translate to better at-home behavior, thus making the parents' jobs easier. Additionally, other sensory rooms and spaces across the world who may be able to learn from the devices in the Whitemarsh Elementary Sensory Space.

Justification:

Sensory Room General Info

- Sensory rooms are designed to be calming, stimulating, or some sort of combination of both. They do this by utilizing components and devices that appeal to the five senses: touch, taste, smell, sight, hearing. They are particularly helpful for people who suffer from sensory-processing disorders and/or cognitive disabilities. Some examples are autism, down-syndrome, dementia, etc. By engaging the senses, sensory rooms help to improve fine-motor skills, cognitive abilities, and also provide a calming environment when it's needed

Items to Have in a Sensory Room

- Lights are an integral part of sensory rooms. Different moods can be created depending on the color of light. Color changing LEDs can be used by non-verbal kids to express

their emotions. Projected images can be used to stimulate the visual senses of the children. Fiber optic strands can be used for multi-sensory effect. The strands are completely safe to touch, easy to set up, and allow the children to interact with the environment.

<https://enablingdevices.com/blog/benefits-sensory-rooms-special-needs/>

- Sensory Rooms are designed to facilitate people who struggle with sensory integration, or the link between the brain and our behavior. Individuals with developmental disorders may struggle with processing sensory experiences in everyday life. In some cases, the brain may overreact to sensory experiences, or not react enough. They can use a sensory room to help learn sensory integration for themselves by working softer and safer sensory experiences, and develop skills which in turn can be

<https://www.procaretherapy.com/blog/sensory-rooms-schools/>

- Sensory rooms in schools can increase concentration during class instruction, provide mental calmness when the student may be over-stimulated or upset, certain items can improve motor skills. Good items to use in sensory rooms are bubble tubes, bean bags,

<https://sensoryprocessinghub.humber.nhs.uk/sense-visual/#:~:text=It%20is%20estimated%20that%2080,reading%2C%20writing%20and%20playing%20sports.>

- About 80-85% of all learning comes from the visual sense. People learn things like reading body language, and spotting danger. The visual sense also helps to boost comprehension and auditory processing. This is why visual stimulation is crucial for development.

Problem Justification:

This problem is important to solve now because of how important visual sensory is to education and development in children, and also because of the calming experience sensory rooms in general can provide to special-needs children. The sensory room at WES is lacking in the visual sensory department, so it's imperative that we design, fabricate, and implement a new device that will serve to stimulate and/or provide a calming experience for the special-needs kids at WES. About 80-85% of learning comes from visual stimulation, with benefits ranging from increased comprehension and auditory development to enhanced ability to read body language and identifying threats. Furthermore, visual sensory can provide a calming environment for kids who may feel overstimulated or just need a place to zone out and escape. Lights, LEDs, and fiber optics are all forms of visual sensory stimulation that can be utilized to provide kids with the numerous benefits of visual stimulation.